3.24 Derivations

Definition 3.24.1 (Module of Derivations)
Let B be a commutative ring and M an (B,C)-bimodule, then we say a derivation is a map $D : B \to M$ satisfying the following properties

- $D(x+y) = D(x) + D(y)$ ***linearity***
- $D(xy) = xD(y) + yD(x)$ ***product rule***

We denote the family of such derivations by $\mathrm{Der}(B,M)$, and it is an (B,C)-bimodule. We may consider the case $M = C$ and $\phi : B \to B$ a ring homomorphism and in this case the product rule becomes

$$D(xy) = \phi(x)D(y) + D(x)\phi(y)$$

We observe that by induction we have $D(n \cdot 1_B) = 0$ for all $n \in \mathbb{Z}$.

Suppose that B is an A-algebra then we claim the following are equivalent

- $D(a) = 0 \quad \forall a \in A$
- $D(ab) = aD(b) \quad \forall a \in A$

In this case we denote the set of A-linear derivations by

$$\mathrm{Der}_A(B,M)$$

Note every derivation is $\mathbb{Z}$-linear so we may work with $A = \mathbb{Z}$ in these cases. Finally we write

$$\mathrm{Der}_A(B) := \mathrm{Der}_A(B,B)$$

For our purposes the following will be the key example

Example 3.24.2 (Evaluation at a zero)
Consider the case $B = A[X_1,...,X_n]/\mathfrak{a}$ a f.g. A-algebra, L an A-algebra and $x \in L^n$ a zero of $\mathfrak{a}$. Then we may define the B-module structure on L by

$$f \cdot \lambda := f(x)\lambda$$

In this case we have the more explicit form

$$\mathrm{Der}_A(B,L;x) = \{D \in \mathrm{Hom}_A(B,L) \mid D(fg) = f(x)D(g) + g(x)D(f)\}$$

which is a (B,L)-bimodule.

Lemma 3.24.3
Suppose $D, D' \in \mathrm{Der}_A(B,M)$ agree on a set of generators for B, then they are identically equal. For example in the previous example when $D\left(\overline{X}_i\right) = D'\left(\overline{X}_i\right)$.

The following is a useful technical tool to identify derivations as algebra homomorphisms

Definition 3.24.4 (Dual Ring)
Let M be a B-module. Define the ***dual ring*** *as follows*

$$B[M] := B \times M$$

$$(b,m) \times (b',m') = (bb', bm' + b'm)$$

with addition defined in the obvious way and multiplicative unit is $(1_B, 0_M)$. Observe that it has an ideal $N := 0 \times M$ such that $N^2 = 0$ and canonically $B[M]/N \cong B$.

When B is an A-algebra, then there is a natural A-algebra structure on $B[M]$.

Proposition 3.24.5
Let M be an B-module where B is an A-algebra then there is a bijection

$$\begin{aligned} \mathrm{Der}_A(B,M) &\longrightarrow \mathrm{AlgHom}_A(B, B[M]) \\ D &\to \quad \phi_D(b) = (b, D(b)) \end{aligned}$$

Definition 3.24.6 (Partial Derivative)
Suppose $F \in A[X_1,...,X_n]$ given by

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$$F(X_1,...,X_n) = \sum_{i\in\mathbb{N}^n} a_i X_1^{i_1}...X_n^{i_n} \quad a_i \in A$$

Define the partial derivative as follows

$$\frac{\partial F}{\partial X_k} := \sum_{\substack{i\in\mathbb{N}^n \\ i_k\neq 0}} a_i i_k X_1^{i_1}...X_k^{i_k-1}...X_n^{i_n}$$

We observe that

$$\frac{\partial X_l}{\partial X_k} := \delta_{lk}$$

Note this condition uniquely determines the equation above, see (3.24.8).

We show that these form a basis for $\mathrm{Der}_k(A)$.

Lemma 3.24.7 (Product Rule)
*For $F,G \in A[X_1,...,X_n]$ we have the **product rule***

$$\frac{\partial FG}{\partial X_k} = F\frac{\partial G}{\partial X_k} + G\frac{\partial F}{\partial X_k}$$

Proof. First we demonstrate the result in the univariate case $n = 1$. For monomials this is straightforward:

$$\frac{\partial X^r X^s}{\partial X} = (r+s)X^{r+s-1} = X^s\frac{\partial X^r}{\partial X} + X^r\frac{\partial X^s}{\partial X}$$

For general univariate polynomials the product rule follows from the linearity of $\partial/\partial X$. The multivariate case then follows from considering the isomorphism $A[X_1,...,X_n] \cong A\left[X_1,...,\widehat{X_k},...,X_n\right][X_k]$ under which $\partial/\partial X_k$ corresponds to $\partial/\partial X$. ∎

Lemma 3.24.8 (Multivariate Chain Rule)
Let $D \in \mathrm{Der}_A(B,M)$ be a derivation, $x_1,...,x_n \in A$ and $F \in B[X_1,...,X_n]$. Then

$$D(F(x_1,...,x_n)) = \sum_{k=1}^{n} \frac{\partial F}{\partial X_k}(x_1,...,x_n)D(x_k)$$

As a special case we find

$$D(F(x)) = F'(x)D(x)$$
$$D(x^n) = nx^{n-1}D(x)$$

Proof. First we observe that by induction on n

$$D(x_1...x_n) = \sum_{k=1}^{n} x_1...\widehat{x_k}...x_n D(x_k)$$

and in particular

$$D(x^n) = \begin{cases} nx^{n-1}D(x) & n > 0 \\ 0 & n = 0 \end{cases}$$

Then for $F \in k[X_1,...,X_n]$ we have

$$\begin{aligned} D(F(x_1,...,x_n)) &= \sum_{i\in\mathbb{N}^n} a_i D\left(x_1^{i_1}...x_n^{i_n}\right) \\ &= \sum_{i\in\mathbb{N}^n} a_i \sum_{k=1}^{n} x_1^{i_1}...\widehat{x_k^{i_k}}...x_n^{i_n} D\left(x_k^{i_k}\right) \\ &= \sum_{k=1}^{n}\sum_{i\in\mathbb{N}^n} a_i x_1^{i_1}...\widehat{x_k^{i_k}}...x_n^{i_n} D\left(x_k^{i_k}\right) \\ &= \sum_{k=1}^{n}\sum_{\substack{i\in\mathbb{N}^n \\ i_k\neq 0}} i_k a_i x_1^{i_1}...x_k^{i_k-1}...x_n^{i_n} D(x_k) \end{aligned}$$

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which gives the required result. ∎

Proposition 3.24.9
Let $B \coloneqq A[X_1, ..., X_n]$ be a polynomial ring and M a (B, C)-bimodule. Then $\mathrm{Der}_A(B, M)$ is a finite free (B, C)-bimodule with basis $\{\partial/\partial X_1, ..., \partial/\partial X_n\}$ (3.24.6).

Proposition 3.24.10 (Extensions to Quotient Field)
Let B be an A-algebra, S a multiplicative set of B and M a B-module. Then we have the following isomorphisms

$$\mathrm{Der}_A(B, M) \xrightarrow{\sim} \mathrm{Der}(S^{-1}B, S^{-1}M)$$
$$D \to \frac{b}{s} \to \frac{bD(s) - sD(b)}{s^2}$$

In particular if $M = \Omega$ is a field containing B (which is therefore an integral domain) then we have an isomorphism

$$\mathrm{Der}_A(B, \Omega) \xrightarrow{\sim} \mathrm{Der}_A(K, \Omega)$$

where $K \coloneqq \mathrm{Frac}(B)$.

In light of (3.24.5) the following definition is clearly related to the existence of derivations.

Definition 3.24.11 (Formally Smooth)
*Let B be an A-algebra. We say it is **formally smooth** if for any A-algebra C with $N \triangleleft C$ such that $N^2 = 0$ and any homomorphism $v : B \to C/N$ there exists a morphism $\overline{v} : A \to C$ making the following diagram commute*

$$\begin{array}{ccc} A & \longrightarrow & C \\ \downarrow & \nearrow & \downarrow \pi \\ B & \xrightarrow{v} & C/N \end{array}$$

*We say that B is **formally unramified** if in addition the lifting is always unique.*

Lemma 3.24.12
Let C be a ring and $N \subseteq \sqrt{(0)}$ an ideal. Then $x \in C$ is invertible if and only if $\overline{x} \in C/N$ is invertible.

Proof. One direction is obvious. Conversely if $\overline{x}$ is invertible then $x \in C^\star + N \subseteq C^\star + \sqrt{(0)} \subseteq C^\star$ by (3.3.68) as required. ∎

Lemma 3.24.13
If a ring B is formally smooth (resp. unramified) then so is $S^{-1}B$.

Proof. It's enough to show that for $s \in S$ we have $\overline{v}(s)$ is invertible, which follows from (3.24.12). ∎

Lemma 3.24.14
The polynomial ring $B \coloneqq A[X_1, ..., X_n]$ and function field $A(X_1, ..., X_n)$ is formally smooth.

Proof. The first follows by the fact $A[X_1, ..., X_n]$ is a free A-algebra and the second by (3.24.13). ∎

Lemma 3.24.15
Let B be a formally smooth (resp. formally unramified) A-algebra and C is a formally smooth (resp. formally unramified) B-algebra. Then C is a formally smooth (resp. formally unramified) A-algebra.

Lemma 3.24.16 (Separable algebraic extensions are "formally unramified")
A separable algebraic extension K/k is formally unramified.

Proof. Consider the following commutative diagram

$$\begin{array}{ccc} k & \longrightarrow & C \\ \downarrow & \nearrow & \downarrow \pi \\ K & \xrightarrow{v} & C/N \end{array}$$

We suppose first that K is a finite extension. By (3.17.108) it is simple, namely $K = k(\alpha) = k[\alpha]$. Let $m(X)$ be the minimal polynomial over k and choose $x \in C$ such that $\pi(x) = v(\alpha)$. Note that $\overline{v}(\alpha) = x$ need not be well-defined because in general $m(x) \neq 0$. We show however there is an $n \in N$ such that $m(x + n) = 0$. For consider any $n \in N$ and $f(X) \in k[X]$.

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$$f(x+n) = f(x) + \sum_{i=0}^{N} \lambda_i [(x+n)^i - x^i]$$
$$= f(x) + f'(x)n$$

as $n^2 = 0$. Therefore we propose $n = -m(x)/m'(x)$; to show this is well-defined, consider firstly $\pi(m(x)) = m(v(\alpha)) = v(m(\alpha)) = 0$ whence $m(x) \in N$. To show that $m'(x)$ is a unit we argue as follows. As α is separable, $m'(\alpha) \neq 0$, hence is a unit in K. As v is injective $v(m'(\alpha)) = m'(v(\alpha)) = m'(\pi(x)) = \pi(m'(x))$ is a unit in C/N. By (3.24.12) $m'(x) \in C^\star$. Therefore $n \in N$ is well-defined and $m(a+n) = 0$ by construction. Therefore the map $\overline{v} : k[\alpha] \to A$ such that $\overline{v}(\alpha) = a + n$ is well-defined, and the diagram commutes because $\pi(a+n) = \pi(a) + \pi(n) = v(\alpha)$.

Suppose we had another $\overline{v}'$, then $a' \coloneqq \overline{v}'(\alpha)$ again satisfies $m(a') = 0$ and $\pi(a') = v(\alpha)$ whence $a' - a \in N$. By the same argument as before

$$m(a') = m(a + (a' - a)) = m(a) + m'(a)(a' - a)$$

As $m'(a) \neq 0$ and $m(a') = m(a) = 0$ we see $a' = a$ which demonstrates uniqueness.

Suppose that K/k is algebraic than for every $\alpha \in K$ we have liftings $v_\alpha : k(\alpha) \to k$. We claim that $v_\alpha|_{k(\alpha)\cap k(\beta)} = v_\beta|_{k(\alpha)\cap k(\beta)}$. However $k(\alpha) \cap k(\beta)$ is also finite and so we are done by uniqueness. ∎

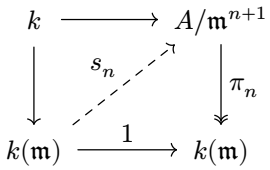
Proposition 3.24.17
A separably generated extension K/k is formally smooth.

Proof. This follows from (3.24.14), (3.24.16) and (3.24.15). ∎

Lemma 3.24.18 (Coefficient Field modulo Powers of the Maximal Ideal)
Let k be a field and $(A, \mathfrak{m})$ a local ring which is a k-algebra, with residue field $k(\mathfrak{m}) \coloneqq A/\mathfrak{m}$.

Write $\rho_{n,m} : A/\mathfrak{m}^{n+1} \to A/\mathfrak{m}^{m+1}$ and $\pi_n : A/\mathfrak{m}^{n+1} \to A/\mathfrak{m} = k(\mathfrak{m})$ for the canonical reduction maps.

If $k(\mathfrak{m})/k$ is separably generated then for every $n \geq 0$ there is a k-algebra homomorphism s_n such that the following diagram commutes

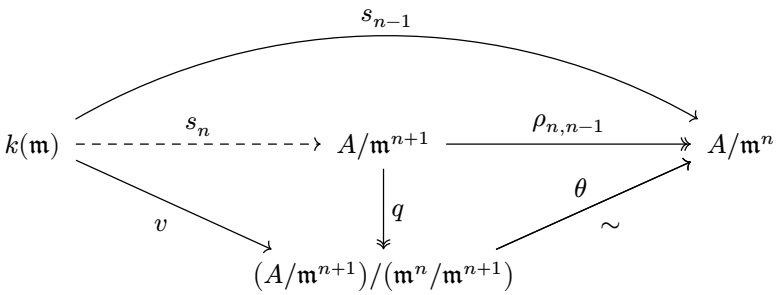


If $k(\mathfrak{m})/k$ is separable algebraic then this morphism is unique. In any case the s_n can be chosen compatibly, $\rho_{n,n-1} \circ s_n = s_{n-1}$, and in the separable algebraic case they are automatically compatible, since $\rho_{n,n-1} \circ s_n$ makes the diagram for $n-1$ commute ($\pi_{n-1} \circ \rho_{n,n-1} = \pi_n$).

If $k(\mathfrak{m}) = k$, that is $k \to k(\mathfrak{m})$ is an isomorphism, all of this is trivial: s_n is the structure map $k \to A/\mathfrak{m}^{n+1}$ composed with the inverse of $k \cong k(\mathfrak{m})$, which is unique because a k-algebra homomorphism out of k is the structure map, and compatible.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

The diagram. We construct s_n by induction on n, the case $n = 0$ being $s_0 = 1_{k(\mathfrak{m})}$ as $A/\mathfrak{m} = k(\mathfrak{m})$. For $n \geq 1$ consider the following diagram, in which q is the quotient map by the ideal $\mathfrak{m}^n/\mathfrak{m}^{n+1}$ of $A/\mathfrak{m}^{n+1}$ (which has square zero, as $2n \geq n+1$), $\rho_{n,n-1}$ is the reduction, which is surjective with kernel $\mathfrak{m}^n/\mathfrak{m}^{n+1}$ (the short exact sequence of the third isomorphism theorem (3.3.106) for $\mathfrak{m}^{n+1} \subseteq \mathfrak{m}^n \subseteq A$), so that by the isomorphism theorem (3.3.57) it induces an isomorphism θ with $\theta \circ q = \rho_{n,n-1}$ (a k-algebra isomorphism, as q and $\rho_{n,n-1}$ are k-algebra homomorphisms), s_{n-1} is given by the induction hypothesis, and $v \coloneqq \theta^{-1} \circ s_{n-1}$:



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Existence. If $k(\mathfrak{m})/k$ is separably generated it is formally smooth (3.24.17), so the lifting property (3.24.11), applied to the square-zero ideal $\mathfrak{m}^n/\mathfrak{m}^{n+1}$ and the homomorphism $v : k(\mathfrak{m}) \to (A/\mathfrak{m}^{n+1})/(\mathfrak{m}^n/\mathfrak{m}^{n+1})$, gives a k-algebra homomorphism $s_n : k(\mathfrak{m}) \to A/\mathfrak{m}^{n+1}$, the dashed arrow, with $q \circ s_n = v$. Hence $\rho_{n,n-1} \circ s_n = \theta \circ q \circ s_n = \theta \circ v = s_{n-1}$, the compatibility.

s_n is a section. As $\pi_n = \pi_{n-1} \circ \rho_{n,n-1}$,

$$\pi_n \circ s_n = \pi_{n-1} \circ \rho_{n,n-1} \circ s_n = \pi_{n-1} \circ s_{n-1} = 1_{k(\mathfrak{m})}$$

by induction, so s_n is a section of π_n.

Uniqueness. Now let $k(\mathfrak{m})/k$ be separable algebraic. It is separably generated (with empty transcendence basis), so the above applies. Moreover it is formally unramified (3.24.16), so the lift s_n of s_{n-1} is unique. We show by induction on n that s_n is the only section of π_n. For $n = 0$, $\pi_0 = 1_{k(\mathfrak{m})}$, so a section t satisfies $t = \pi_0 \circ t = 1_{k(\mathfrak{m})}$. For $n \geq 1$, if t is a section of π_n then $\rho_{n,n-1} \circ t$ is a section of π_{n-1} (as $\pi_{n-1} \circ \rho_{n,n-1} = \pi_n$), hence equals s_{n-1} by induction, so t is a lift of s_{n-1}, that is $t = s_n$. In particular the s_n are automatically compatible. ∎

Remark 3.24.19

A compatible family $(s_n)_{n\geq 0}$ is the same as a single k-algebra homomorphism $s : k(\mathfrak{m}) \to \hat{A} := \varprojlim_n A/\mathfrak{m}^{n+1}$ with $\pi \circ s = 1_{k(\mathfrak{m})}$, by the universal property of the inverse limit. So when $k(\mathfrak{m})/k$ is separably generated the residue field lifts to a subfield of the completion of A. This is a coefficient field in the sense of Cohen. It is unique when $k(\mathfrak{m})/k$ is separable algebraic, but in general it is not, and two choices differ by a k-derivation of $k(\mathfrak{m})$.

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